Finding the Error: Correspondence and Scale Factor Answer Key

High School Geometry

Quick Answers

1. (a) EF = 12 cm, —
2. (a) TU = 6 in, —
3. (a) scale factor = 1.5, (b) YZ = 30 m, —
4. (a) angle P = 65, —
5. (a) EF = 10.5 cm, —
6. (a) x = 4, —

What is verified

0 of 6 problems fully machine-checked against SymPy.
— marks an answer only you can judge — problems 1, 2, 3, 4, 5, 6. The worked solution states what a correct response must contain.

Curriculum

Level: High School Geometry
HSG-SRT.A.2 – *problems 3, 4*
HSG-SRT.B.5 – *problems 1, 2, 5, 6*
Difficulty 2–4 of 5 across 13 tagged checks.

Common wrong answers

1. *If they got 13.5: reused the 9 cm side in place of the 8 cm side, solving 6/9 = 9/EF*
2. *If they got 37.5: turned the second ratio upside down, solving 10/4 = TU/15*
3. *If they got 13.33: used the reciprocal scale factor and shrank the side instead of enlarging it*
4. *If they got 75: matched P with K instead of L and copied the 75 degree angle*
5. *If they got 168: multiplied the full-size side by 4 instead of dividing by it*
6. *If they got 14: wrote the second ratio upside down as 15/10*

1. $\triangle ABC \sim \triangle DEF$ with $AB = 6$ cm, $BC = 8$ cm, and $DE = 9$ cm. The student wrote $\frac{6}{9} = \frac{9}{EF}$ and got $EF = 13.5$ cm. (a) Correct length of $\overline{EF}$. (b) Name the error.
Solution (a). Corresponding sides are $\overline{AB} \leftrightarrow \overline{DE}$ and $\overline{BC} \leftrightarrow \overline{EF}$, so each ratio must compare one side of $\triangle ABC$ with the matching side of $\triangle DEF$:

$$\frac{AB}{DE} = \frac{BC}{EF} \qquad \frac{6}{9} = \frac{8}{EF}.$$

Cross-multiplying, $6 \cdot EF = 8 \cdot 9 = 72$, so $\boxed{EF = 12 \text{ cm}}$

Solution (b) — instructor-judged. A complete response says the second ratio used the 9 cm side again in place of the 8 cm side of $\triangle ABC$, so both numerators came from the wrong triangles. Each ratio must pair one side of the first triangle with its matching side in the second. Award full credit only when the mismatched pair is named.

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2. $\triangle PQR \sim \triangle STU$ with $PQ = 10$ in, $QR = 15$ in, and $ST = 4$ in. The student wrote $\frac{10}{4} = \frac{TU}{15}$ and got $TU = 37.5$ in. (a) Correct length of $\overline{TU}$. (b) Name the error.
Solution (a). With $\overline{PQ} \leftrightarrow \overline{ST}$ and $\overline{QR} \leftrightarrow \overline{TU}$,

$$\frac{10}{4} = \frac{15}{TU} \qquad \Rightarrow \qquad 10 \cdot TU = 60.$$

Therefore $\boxed{TU = 6 \text{ in}}$

Solution (b) — instructor-judged. A complete response says the second ratio was inverted: TU belongs to the smaller triangle $\triangle STU$, so it must sit in the denominator, where 4 sits on the left side. Writing $\frac{TU}{15}$ puts the small triangle on top on one side and on the bottom on the other. A response that only recomputes without naming the inversion earns partial credit.

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3. $\triangle ABC \sim \triangle XYZ$ with $AB = 12$ m, $BC = 20$ m, and $XY = 18$ m. The student used scale factor $\frac{12}{18}$ and got $YZ = 13.33$ m. (a) Correct scale factor. (b) Correct YZ. (c) Explain the error.
Solution (a). The factor that takes $\triangle ABC$ to $\triangle XYZ$ multiplies each side of $\triangle ABC$ by the ratio of the image length to the original length:

$$k = \frac{XY}{AB} = \frac{18}{12}.$$

So $\boxed{k = 1.5}$

Solution (b). Multiply the matching side by k: $YZ = 20 \cdot 1.5$, so $\boxed{YZ = 30 \text{ m}}$

Solution (c) — instructor-judged. A complete response says the factor was written upside down. Going from the smaller triangle to the larger one the factor is the larger length over the smaller, so it must exceed 1; a factor below 1 shrinks a side that should grow, which is why 13.33 m came out shorter than the 20 m side it was scaled from. Full credit requires the direction to be stated, not just the corrected arithmetic.

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4. $\triangle JKL \sim \triangle MNP$ with $m\angle J = 40°$ and $m\angle K = 75°$. The student concluded $m\angle P = 75°$. (a) Correct $m\angle P$. (b) Name the error.
Solution (a). The statement pairs J with M, K with N, and L with P, so $m\angle P = m\angle L$. The angles of $\triangle JKL$ sum to 180°:

$$m\angle L = 180 - 40 - 75.$$

Therefore $\boxed{m\angle P = 65°}$

Solution (b) — instructor-judged. A complete response says P is the third vertex of $\triangle MNP$, so it corresponds to L, the third vertex of $\triangle JKL$ — not to K, whose partner is

N. The copied 75° is the measure of the second vertex pair. Award full credit only when the correct pairing is named.

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5. A poster shop reduces a triangular decal: $\triangle ABC$ has $AB = 30$ cm and $BC = 42$ cm, and the reduced copy $\triangle DEF$ has $DE = 7.5$ cm. The student wrote $EF = 42 \cdot \frac{30}{7.5} = 168$ cm. (a) Correct length of $\overline{EF}$. (b) Explain the direction error.

Solution (a). From original to copy the factor is $\frac{DE}{AB} = \frac{7.5}{30} = \frac{1}{4}$, so

$$EF = 42 \cdot \frac{7.5}{30}.$$

That gives $\boxed{EF = 10.5 \text{ cm}}$

Solution (b) — instructor-judged. A complete response says the factor the student used runs from the copy to the original, so travelling the other way the side must be divided by it rather than multiplied. The giveaway is that 168 cm is longer than the 42 cm side it came from, even though the copy is a reduction. Full credit requires the direction to be stated in words.

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6. $\triangle ABC \sim \triangle DEF$ with $AB = (x + 4)$ cm, $DE = 12$ cm, $BC = 10$ cm, and $EF = 15$ cm. The student wrote $\frac{x+4}{12} = \frac{15}{10}$ and got $x = 14$. (a) Solve correctly for x. (b) Name the error.

Solution (a). Both ratios must read first triangle over second triangle:

$$\frac{AB}{DE} = \frac{BC}{EF} \qquad \frac{x+4}{12} = \frac{10}{15} = \frac{2}{3}.$$

So $x + 4 = 12 \cdot \frac{2}{3} = 8$, giving $\boxed{x = 4}$

Solution (b) — instructor-judged. A complete response says the right-hand ratio was written with its sides swapped: $\overline{BC}$ belongs to the same triangle as $\overline{AB}$, so it must sit on top, with $\overline{EF}$ underneath. Reversing it makes the right side larger than 1 and inflates x. Full credit requires naming the swap, not just producing the right number.